

Trump Occurrence Forecasting Benchmark

Offline probabilistic forecasting of future word and phrase occurrence in chronologically ordered public transcripts.

STOP

Locked-test decision

The content-feature challenger passed validation, then increased Brier error by **0.001115 (2.58%)** against the timing/survival baseline in the single locked chronological test pass.

BENCHMARK AT A GLANCE

174

transcripts

180

frozen targets

38,790

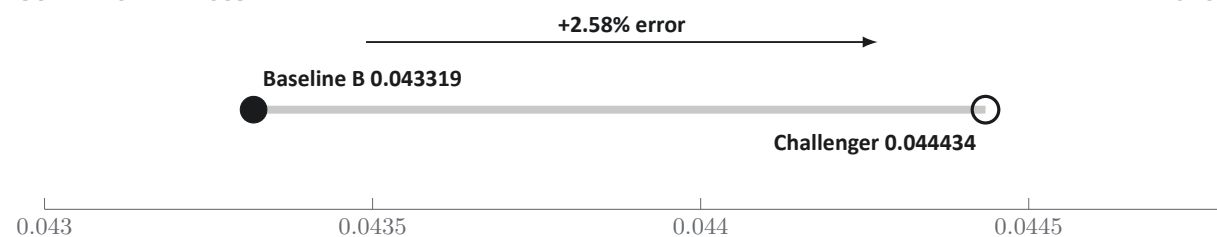
locked-test rows

API-free

compute profile

LOCKED-TEST BRIER SCORE

Lower is better



Bootstrap interval for challenger minus Baseline B: [0.000259, 0.002074]. The interval remains entirely above zero.

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PROTOCOL

Single locked test pass

REPRODUCIBLE ARTIFACT

trump-occurrence-sprint-handoff-2026-06-15.zip

Executive Summary

This benchmark evaluates whether transcript-content features improve forecasts of whether a target word or phrase will appear later in a Rev transcript. The challenger augments a timing/survival baseline with train-only target-adjacent content features, a fixed duration-sensitivity term, and validation-fitted calibration. The evaluation uses chronological transcript-level splits and a single locked test pass.

Primary conclusion

Baseline B generalized better. Its locked-test Brier score was **0.043319**, compared with **0.044434** for the full challenger. The challenger therefore increased Brier error by **0.001115 (2.58% relative)**. A 1,000-replicate bootstrap interval for challenger minus baseline Brier was **[0.000259, 0.002074]**, entirely above zero.

The negative result is consistent across the primary metric, after-start rows, and the content ablation. Blanking the prefix produced Brier 0.044460, nearly identical to the full challenger and still worse than Baseline B. The protocol therefore returns **STOP** for this challenger path.

The result is operationally informative rather than merely null. It establishes that, under this corpus size and temporal shift, local TF-IDF/SVD and adjacency features do not add reliable out-of-time signal beyond timing and survival. A future continuation would require a materially different model class or a new corpus and a newly locked test, rather than tuning against the exposed test set.

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1 Objective and Task Definition

1.1 Forecasting task

For transcript i , target j , and checkpoint t , the benchmark predicts the probability that target j will occur later in the transcript, conditional on the target not having appeared by checkpoint t :

$$p_{ijt} = \Pr(Y_{ijt} = 1 \mid \text{target unseen through checkpoint } t).$$

The outcome Y_{ijt} is one if the target occurs after the checkpoint and zero otherwise. Checkpoints are replayed over held-out transcripts, yielding multiple forecast rows per transcript-target pair.

1.2 Primary decision metric

The primary metric is mean Brier score:

$$\text{Brier} = \frac{1}{N} \sum_{n=1}^N (p_n - y_n)^2.$$

Lower values indicate better probabilistic accuracy. Expected calibration error (ECE), log loss, and AUC are reported as diagnostics. They do not override the predeclared Brier-based decision rule.

1.3 Decision rule

The challenger had to pass validation before the locked test could be opened. After validation, the test set was evaluated once. Failure to improve on Baseline B on the full locked-test rows triggers **STOP**.

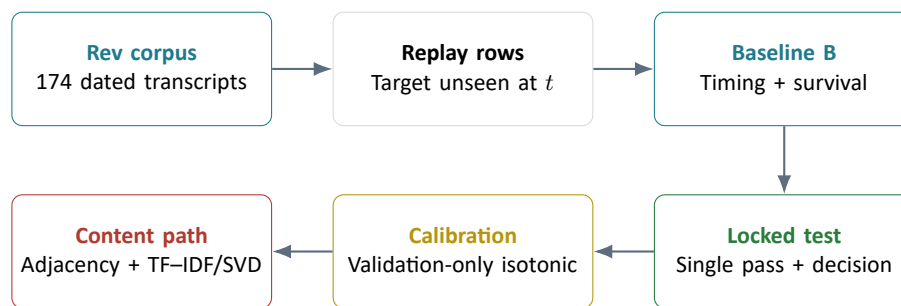


Figure 1: Benchmark execution path and locked-test control.

2 Dataset and Experimental Design

2.1 Corpus construction

The final corpus contains 174 dated Donald Trump transcripts discovered from Rev’s Donald Trump category page and Rev’s own sitemap. Raw HTML was saved locally. Cleaning preferred Donald Trump speaker turns when labeled, otherwise selected a dominant speaker when reliable, and fell back to all parsed turns when necessary. Exact normalized text hashes and repeated title/date pairs were deduplicated.

Table 1: Corpus and benchmark scale.

Item	Value
Transcript count	174 retained transcripts
Corpus date range	July 5, 2019 to June 7, 2026
Transcript length	Median 3,041 words; interquartile range 1,210–6,973
Frozen targets	180 train-only words or phrases
Target bands	40 high, 40 mid, 40 rare, 30 cold, 30 phrase
Locked-test rows	38,790 forecast rows across 26 transcripts

2.2 Chronological split

The split is chronological and transcript-level: no transcript appears in more than one split. The validation and test periods are dominated by newer material, which creates a meaningful temporal generalization test.

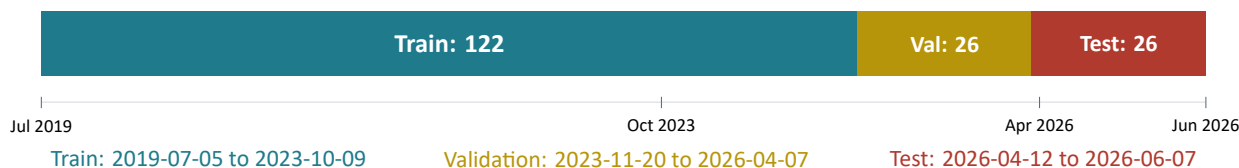


Figure 2: Chronological transcript-level split. Bar width reflects transcript count; dates are annotated below.

Table 2: Format distribution by split.

Split	Interview	Legal	Press	Rally	Speech	Unknown	Total
Train	10	4	32	16	23	37	122
Validation	0	1	4	4	1	16	26
Locked test	1	0	7	0	1	17	26

2.3 Leakage and freeze controls

- Targets were selected from training data only and frozen before validation and test.
- Content features and adjacency statistics were fit using training data only.
- Calibration was fit on validation predictions and frozen before the test pass.
- Model, corpus, split, target, configuration, and validation artifacts were hashed before test-label use.
- The locked test script was run once for the reported decision.

3 Models and Validation Gate

Table 3: Compared forecasting systems.

System	Definition
Baseline A: constant	Forecasts from a fixed occurrence-rate reference.
Baseline B: timing/survival	Uses checkpoint timing, exposure, and target survival information without transcript-content features.
Challenger full	Adds train-only target-adjacent features, trailing and full-prefix TF-IDF/SVD representations, and fixed duration sensitivity over Baseline B in logit space; calibrated on validation.
Challenger blanked	Runs the challenger path with content blanked to estimate whether transcript content contributes useful incremental signal.

The challenger passed the validation gate: Baseline B validation Brier was **0.040780**, while challenger Brier was **0.040337**. Duration sensitivity was directionally valid, with monotonic fraction **0.766** and mean high-minus-low sensitivity **0.074876**. These checks authorized the single locked-test pass.

Why LoRA was not trained

LoRA was a P1 extension, not the P0 acceptance path. The feature challenger completed the required validation and locked-test protocol and produced a valid STOP decision. Training an additional model after opening the locked test would not change that result and would risk post-test model selection.

4 Locked-Test Results

4.1 Primary metrics

Table 4: Locked-test performance on 38,790 rows. Lower is better for Brier, ECE, and log loss.

Model	Brier	ECE	Log loss	AUC
Constant baseline	0.048 416	0.022 703	0.170 778	0.923 560
Timing/survival baseline	0.043 319	0.018 365	0.150 796	0.946 593
Challenger full	0.044 434	0.018 069	0.154 722	0.943 082
Challenger blanked	0.044 460	0.014 503	0.155 044	0.941 189

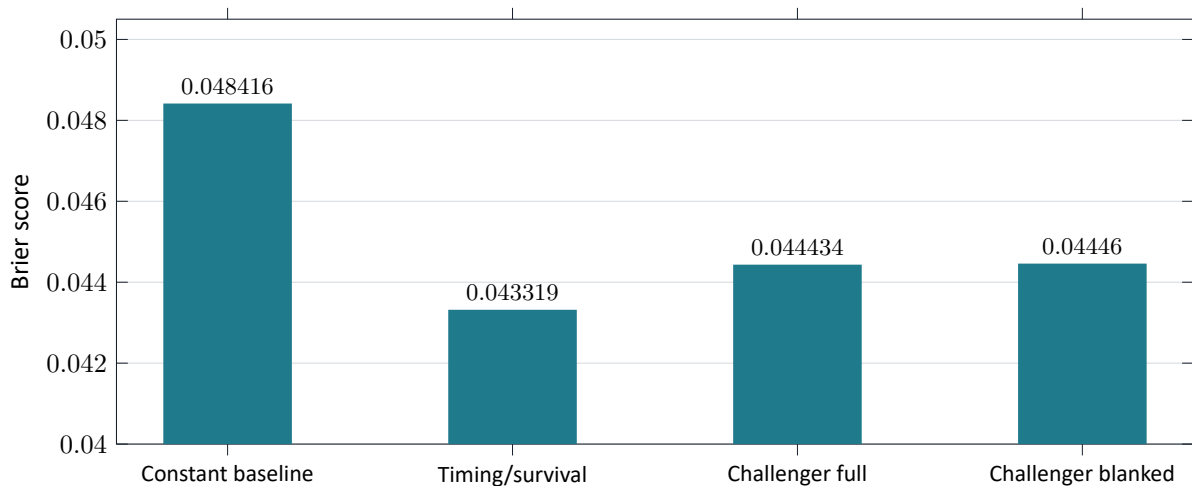


Figure 3: Locked-test Brier score. Baseline B is the best-performing system.

4.2 Decision uncertainty

The observed challenger-minus-baseline Brier delta is $\Delta = +0.001115$. Positive values are unfavorable because they indicate greater error than Baseline B.

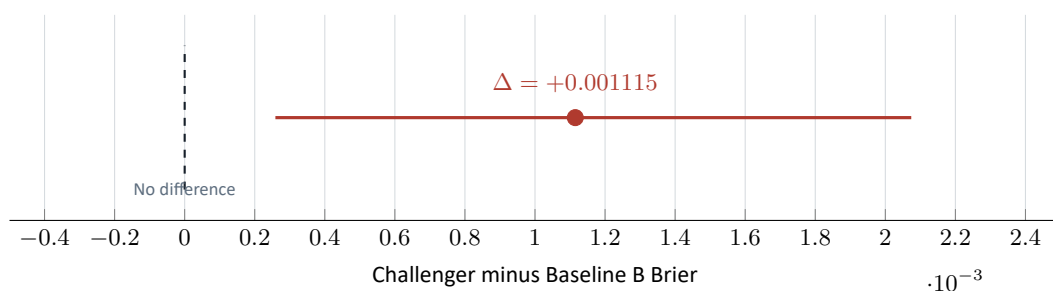


Figure 4: Bootstrap 95% interval from 1,000 transcript-level replicates. The entire interval is above zero.

Decision interpretation. Baseline B outperformed the challenger on the primary metric. The bootstrap interval does not include zero, and the direction is unfavorable throughout. The first STOP condition is therefore met.

5 Calibration and Ablation Diagnostics

5.1 Calibration and loss profile

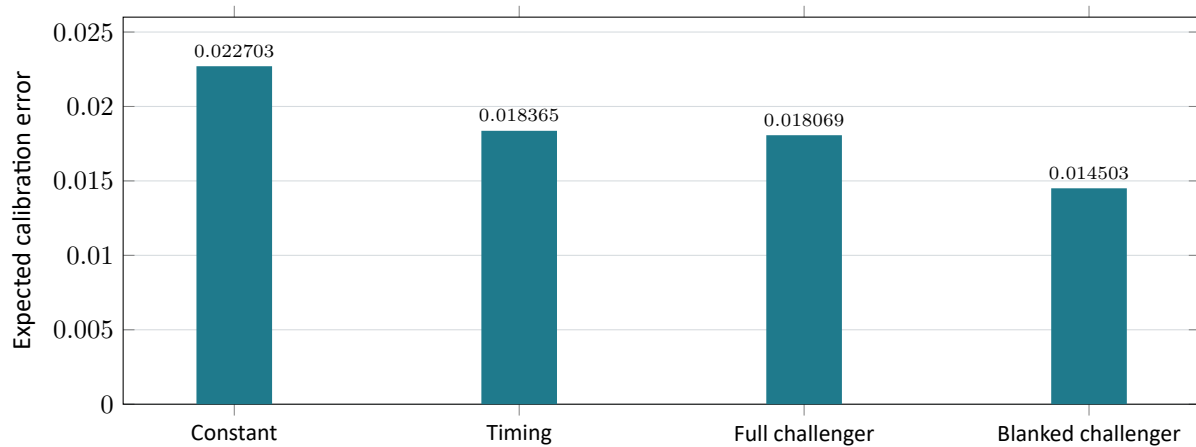


Figure 5: Expected calibration error. Calibration is acceptable but does not offset worse Brier performance.

5.2 Reliability curves

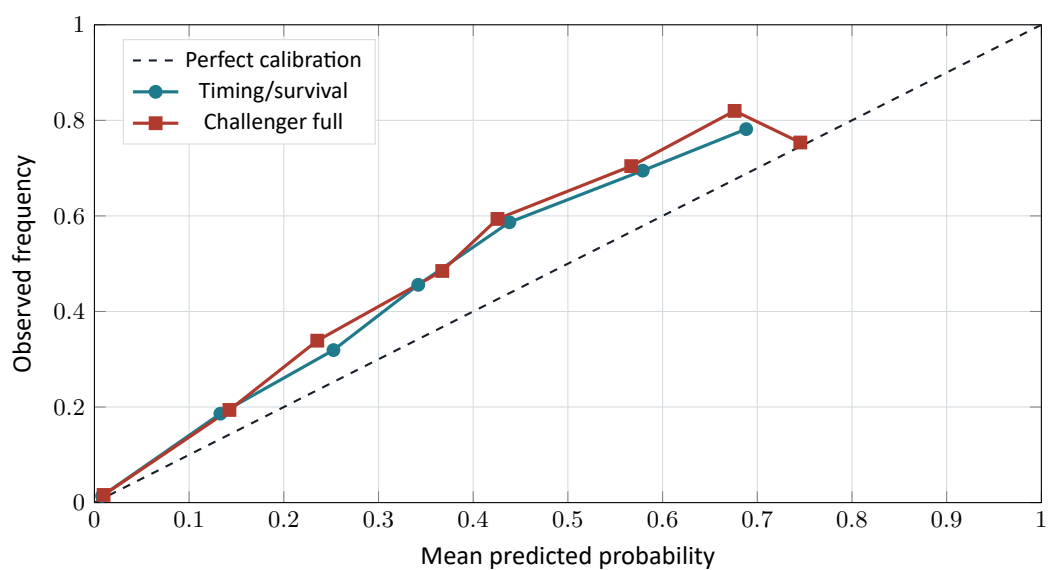


Figure 6: Reliability curves using populated probability bins. The extreme challenger bin with seven observations is omitted from the plot but retained in the metric calculation.

5.3 Content ablation

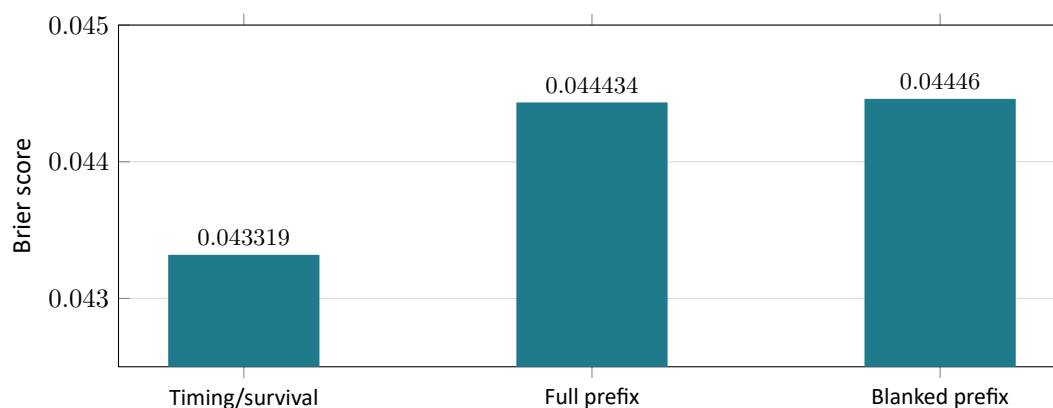


Figure 7: Content ablation. Full-prefix and blanked-prefix variants are nearly identical and both trail Baseline B.

The ablation does not support a useful test-time content contribution. On rows with $t > 0$, Baseline B achieved Brier **0.039284**, while the full challenger achieved **0.040592**. Thus, the challenger remains worse after the transcript has begun, not only at the initial checkpoint.

6 Decision, Limitations, and Next-Step Boundary

6.1 Evidence-to-decision matrix

Table 5: Predeclared decision evidence.

Check	Evidence	Interpretation
Primary Brier	Challenger 0.044434 vs Baseline B 0.043319	Fails continuation bar
Bootstrap interval	Challenger-minus-baseline CI [0.000259, 0.002074]	Entirely unfavorable
Content ablation	Full 0.044434; blanked 0.044460	No useful content lift
After-start rows	Baseline 0.039284; challenger 0.040592	Supports STOP
Calibration	Challenger ECE 0.018069 vs baseline 0.018365	Similar; cannot override Brier
Cost and latency	\$0 API cost; 0.000890 s per validation prediction after feature construction	Operationally feasible

6.2 Limitations

- **Temporal shift.** The locked test is concentrated in newer, shorter 2026 press/event transcripts. This is the intended out-of-time challenge but may penalize train-only lexical features.
- **Speaker extraction.** Primary-speaker extraction is heuristic when Rev labels are missing or ambiguous.
- **Content model class.** The tested challenger is TF-IDF/SVD plus target-adjacent statistics, not a fine-tuned language model.
- **Contamination probe.** A deterministic local distilgpt2 screen flagged zero test transcripts. This is not proof that no public transcript occurs in a larger model’s pretraining data.
- **Single test pass.** The test set is now exposed and cannot be used for further model selection without invalidating the original protocol.

6.3 Bounded next step

The appropriate conclusion is to stop this specific challenger path. A new attempt would require one of the following before any new locked evaluation:

1. a materially larger or more diverse transcript corpus;
2. a different representation or model class with stronger temporal transfer;

3. a new target construction strategy; and
4. a newly reserved chronological test period.

Tuning against the current test outcomes is not a valid continuation.

7 Reproducibility Record

Table 6: Primary package records.

Artifact	Record
Handoff package	trump-occurrence-sprint-handoff-2026-06-15.zip
Package SHA256	B7DC0E2D84241AC7716286313807753B88C54EFE295B19808D02DD04C5E4EC0E
Final decision	DECISION.md
Primary test metrics	metrics/test_metrics.json
Bootstrap results	metrics/bootstrap_deltas.json
Freeze manifest	metrics/freeze_manifest.json
Data card	data/data_card.md
Model artifacts	models/baselines.pkl, models/challenger.pkl

7.1 Reproduction sequence

```
py -3 -m pip install -r requirements.txt
py -3 -m pytest tests
py -3 -m src.run_phase0
py -3 -m src.run_phase1
py -3 -m src.run_phase2
py -3 -m src.run_phase3
py -3 -m src.run_phase4_locked_test
```

Protocol warning. The final command is the single locked-test pass. It should not be rerun for model selection after inspecting the reported test metrics.

7.2 Integrity statement

The handoff archive contains 245 entries, including 180 saved raw HTML artifacts. The archive checksum matches its sidecar file. Frozen model, configuration, corpus, split, target, and validation artifacts match the hashes recorded in the freeze manifest. The sprint test suite passes 11 tests.

End of technical report